# Snubbers in Paralleled IGBT Modules

Application Notes for the ABB/Hitachi 5SNA 1000G650300 (6.5 kV / 1000 A HiPak)

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**Abstract**

These notes discuss the role of snubber networks when high-voltage IGBT modules are operated in parallel, with reference to the ABB/Hitachi 5SNA 1000G650300 HiPak module (6.5 kV, 1000 A, SPT$^{++}$ chip-set). The main conclusion is that snubbers are useful in a paralleled configuration, but for a specific reason: they act primarily as *local decoupling elements* that shorten and equalize the commutation loops of the individual modules, thereby limiting turn-off overvoltage and improving dynamic current sharing. They do not correct static current imbalance, which must be addressed by device matching, gate-drive design, and layout. Sizing guidelines and a worked numerical example are given.

## 1 Scope

Paralleling of high-power IGBT modules is common practice when the required phase current exceeds the rating of a single module. This document addresses one specific design question: whether, and how, snubber networks should be used in a paralleled configuration. The quantitative data refer to the 5SNA 1000G650300 [1], but the methodology applies to any high-voltage HiPak-class module. The general sizing procedure (voltage budget, loop energy) follows the approach of the companion snubber design guideline [5].

## 2 Device Overview

Table 1 summarizes the datasheet values relevant to the present discussion [1]. Two figures deserve attention. First, the turn-off is intentionally slow ($R_{G,\mathrm{off}} = 15\,\Omega$, $t_f \approx 380$–$500\,\mathrm{ns}$), which limits the turn-off $di/dt$ of the IGBT itself. Second, the anti-parallel diode exhibits a large reverse recovery ($I_{rr}$ up to 2.49 kA, $Q_{rr}$ up to 2.26 mC at 150 °C); at low temperature and low forward current the recovery is snappier, and this is normally the most violent transient in the commutation loop.

A package-level remark: the module exposes *three* collector–emitter terminal pairs that must be paralleled by the busbar. The module is therefore already an internal parallel connection closed externally, and the busbar symmetry discipline required between modules is equally required across the terminal pairs of a single module.

## 3 The Commutation Loop with Paralleled Modules

With $N$ modules in parallel, the commutation loop splits into a *common* section (DC-link capacitor bank, shared laminated busbar) and $N$ *branch* sections (module terminals, local busbar tabs). Two effects make the paralleled case harder than the single-module case:

1. **Summed $di/dt$ in the common section.** Each module turning off 1000 A in $t_f \approx 0.5\,\mu\mathrm{s}$ contributes roughly 2–2.5 kA/µs. The common busbar section therefore sees $N$ times this

Table 1: Selected 5SNA 1000G650300 data (typical unless noted) [1].

| Parameter | Symbol | Value | Conditions |
|---|---|---|---|
| Blocking voltage | $V_{CES}$ | 6500 V | $T_{vj} \geq 25\,^{\circ}\mathrm{C}$ |
| DC collector current | $I_C$ | 1000 A | $T_C = 110\,^{\circ}\mathrm{C}$ |
| Fall time | $t_f$ | 380–500 ns | 3600 V, 1000 A |
| Turn-off delay | $t_{d(\mathrm{off})}$ | 5.0–5.9 µs | $R_G = 15\,\Omega$ |
| Diode peak recovery current | $I_{rr}$ | 1.71–2.49 kA | 25–150 °C |
| Recovered charge | $Q_{rr}$ | 1.21–2.26 mC | 25–150 °C |
| Short-circuit current | $I_{SC}$ | 4800 A | $V_{CC} = 4500$ V, $t_{psc} \leq 10$ µs |
| Internal stray inductance | $L_{\sigma,\mathrm{int}}$ | 18 nH | CE terminals |
| RBSOA limit | $V_{CEM,\mathrm{chip}}$ | $\leq 6500$ V | at chip level |
| External gate–emitter capacitor | $C_{GE}$ | 220 nF | datasheet test condition |

slope: 4–5 kA/µs for two modules. The inductive overvoltage of the common section scales accordingly.

2. **Asymmetry.** Any inequality between branch inductances produces unequal transient overvoltages and unequal dynamic current sharing during turn-off and, especially, during diode reverse recovery. The module with the lower-inductance branch takes more of the commutated current and sees a different voltage trajectory.

# 4 What the Snubber Does — and Does Not Do

## 4.1 Decoupling function

A low-inductance film capacitor mounted directly across the DC terminals of *each* module (or each half-bridge leg, for side-by-side legs) makes the effective commutation loop of every module short and, if the mounting is identical, equal for all modules. The consequences are:

- the summed $di/dt$ is confined to the short local loop, and the common busbar inductance is largely removed from the switching transient;
- all modules commutate against the same local impedance, which improves dynamic current sharing at turn-off and during diode recovery;
- the turn-off overvoltage at the chip is reduced.

This decoupling function is the primary reason to use snubbers in a paralleled configuration, and it is the arrangement recommended by the manufacturer's paralleling application note [2].

## 4.2 Clamping function

If a hard limit on the peak voltage is required — in particular for short-circuit turn-off, where the interrupted current is $I_{SC} = 4800$ A per module and the loop overvoltage scales accordingly — an RCD clamp per module is preferred over a plain capacitor. The clamp absorbs the loop energy only above the DC-link voltage and does not participate in the normal resonant exchange.

## 4.3 What snubbers do not solve

Snubbers act on the power loop only. They do not correct:

- **Static current sharing.** This is governed by $V_{CE,\mathrm{sat}}$ matching (same production batch) and thermal coupling (common heatsink). The SPT$^{++}$ chip-set has a positive temperature coefficient of $V_{CE,\mathrm{sat}}$ at rated current, which is inherently self-balancing.
- **Gate-side behavior.** Each module requires an individual gate resistor on its own gate lead — a shared gate resistor invites parasitic oscillations between the paralleled gates — plus the external $C_{GE} = 220\,\mathrm{nF}$ per module as in the datasheet test conditions, and a symmetric gate-loop layout with matched propagation delays.

# 5 Chip-Level Voltage Budget

The RBSOA limit is specified at chip level. The chip voltage exceeds the terminal voltage by the drop across the internal stray inductance:

$$V_{CE,\mathrm{chip}} = V_{CE,\mathrm{term}} + L_{\sigma,\mathrm{int}}\,\frac{di}{dt}, \qquad L_{\sigma,\mathrm{int}} = 18\,\mathrm{nH}. \tag{1}$$

The total peak chip voltage during turn-off is

$$V_{CE,\mathrm{chip}} = V_{DC} + \left(L_{\mathrm{common}}\,N + L_{\mathrm{branch}} + L_{\sigma,\mathrm{int}}\right)\left.\frac{di}{dt}\right|_{\mathrm{module}} + \Delta V_{\mathrm{snap}}, \tag{2}$$

where $\Delta V_{\mathrm{snap}}$ accounts for the diode snap-off contribution and $L_{\mathrm{common}}$ is referred to a single module (its $di/dt$ is multiplied by $N$ because it carries the summed current).

One caution on the available margin: 6.5 kV modules are typically operated from a 3.6–4 kV DC link primarily because of cosmic-ray failure-rate derating [4], not to leave room for overshoot. The margin between $V_{DC}$ and $V_{CEM,\mathrm{chip}}$ is therefore not entirely available as an overshoot allowance; the design budget should be set deliberately, typically reserving the larger share for the short-circuit turn-off case.

# 6 Sizing Guidelines

## 6.1 Decoupling capacitor

The capacitor must absorb the magnetic energy of the loop section it decouples with an acceptable voltage excursion $\Delta V$. From $\frac{1}{2}LI^2 = \frac{1}{2}C\,\Delta V^2$:

$$C \;\geq\; L_{\mathrm{loop}}\left(\frac{I_{\mathrm{off}}}{\Delta V}\right)^2. \tag{3}$$

The capacitor must be a low-inductance film type (dry PP), mounted directly on the module terminals with identical hardware on every module; its own ESL adds to $L_{\mathrm{branch}}$ and must be included in Eq. (2).

## 6.2 Worked example

Two modules, $V_{DC} = 3600\,\mathrm{V}$, per-module turn-off current 1000 A, $t_f = 0.5\,\mu\mathrm{s}$ (hence $di/dt \approx 2\,\mathrm{kA/\mu s}$ per module), $L_{\mathrm{common}} = 100\,\mathrm{nH}$, $L_{\mathrm{branch}} = 50\,\mathrm{nH}$.

Without snubbers, Eq. (2) gives

$$\Delta V = (2\cdot 100 + 50 + 18)\,\mathrm{nH} \times 2\,\mathrm{kA/\mu s} \approx 536\,\mathrm{V}, \tag{4}$$

i.e. a peak chip voltage of about 4.14 kV — acceptable in nominal operation, but the same loop interrupting $I_{SC} = 4800\,\mathrm{A}$ per module with a faster current collapse leaves considerably less margin, and the diode snap adds on top.

With a decoupling capacitor per module, the common 100 nH is removed from the transient. Allocating $\Delta V = 300\,\mathrm{V}$ to the decoupled section at $I_{\mathrm{off}} = 1000\,\mathrm{A}$, Eq. (3) yields

$$C \;\geq\; 100\,\mathrm{nH} \times \left(\frac{1000\,\mathrm{A}}{300\,\mathrm{V}}\right)^2 \approx 1.1\,\mu\mathrm{F}. \tag{5}$$

Capacitors in the 1–4 µF class per module are typical for this module family; a larger value relaxes $\Delta V$ at the cost of heavier inter-module resonances (next section).

### 6.3 RCD clamp

If clamping is required, the clamp capacitor is sized with Eq. (3) using the short-circuit current and the clamp voltage rise as $\Delta V$; the discharge resistor is chosen so that $\tau = RC$ is a small fraction of the switching period, and its power rating from $P = \frac{1}{2} C\,\Delta V^2 f_{sw}$ per clamping event rate. The clamp diode must be a fast soft-recovery type rated for the full DC-link voltage.

## 7 Inter-Module Resonances

Plain capacitors on paralleled modules create LC meshes from module to module: the snubber capacitors of adjacent modules resonate through the busbar inductance between them. If the capacitors, their ESL, or their mounting are not identical, the result is ringing and circulating currents flowing through the snubbers after every commutation. Countermeasures, in order of preference:

1. strictly symmetric mechanical layout (identical capacitors, identical mounting, mirror-image busbar);
2. damping: an RC snubber instead of a plain C, with $R \approx \sqrt{L_{\mathrm{mesh}}/C}$ as a near-critical starting value, verified by simulation or measurement;
3. an RCD clamp, which is inherently non-resonant in normal operation.

The dissipation of a damped snubber must be checked: for an RC snubber the resistor processes approximately $CV^2 f_{sw}$ per charge/discharge cycle pair.

## 8 Summary

- Snubbers are useful when paralleling 5SNA 1000G650300 modules, primarily as per-module decoupling capacitors that shorten and equalize the commutation loops.
- The dimensioning transient is normally the diode reverse recovery (and, for clamps, short-circuit turn-off), not the nominal IGBT turn-off, which is slow by design.
- The voltage budget must be evaluated at chip level, and the cosmic-ray derating means the apparent margin to 6.5 kV is not freely spendable on overshoot.
- Snubbers do not fix static sharing or gate-side interactions: individual gate resistors, per-module $C_{GE}$, batch-matched devices, common heatsink, and symmetric layout remain mandatory.
- Identical snubbers with identical mounting on every module; otherwise the snubbers themselves become a source of asymmetry and inter-module ringing.

## References


[1] ABB Power Grids / Hitachi, "5SNA 1000G650300 HiPak IGBT module," Datasheet 5SYA 1465-03, Oct. 2020.

[2] ABB Switzerland Ltd, Semiconductors, "Paralleling of IGBT modules," Application Note 5SYA 2098.

[3] ABB Switzerland Ltd, Semiconductors, "Applying IGBT," Application Note 5SYA 2053.

[4] ABB Switzerland Ltd, Semiconductors, "Failure rates of IGBT modules due to cosmic rays," Application Note 5SYA 2042.

[5] D. Bagnara, "Snubber design guidelines for a high-current IGBT half-bridge leg," internal technical note.